

# Team, Project, Development Plan, and Requirements.

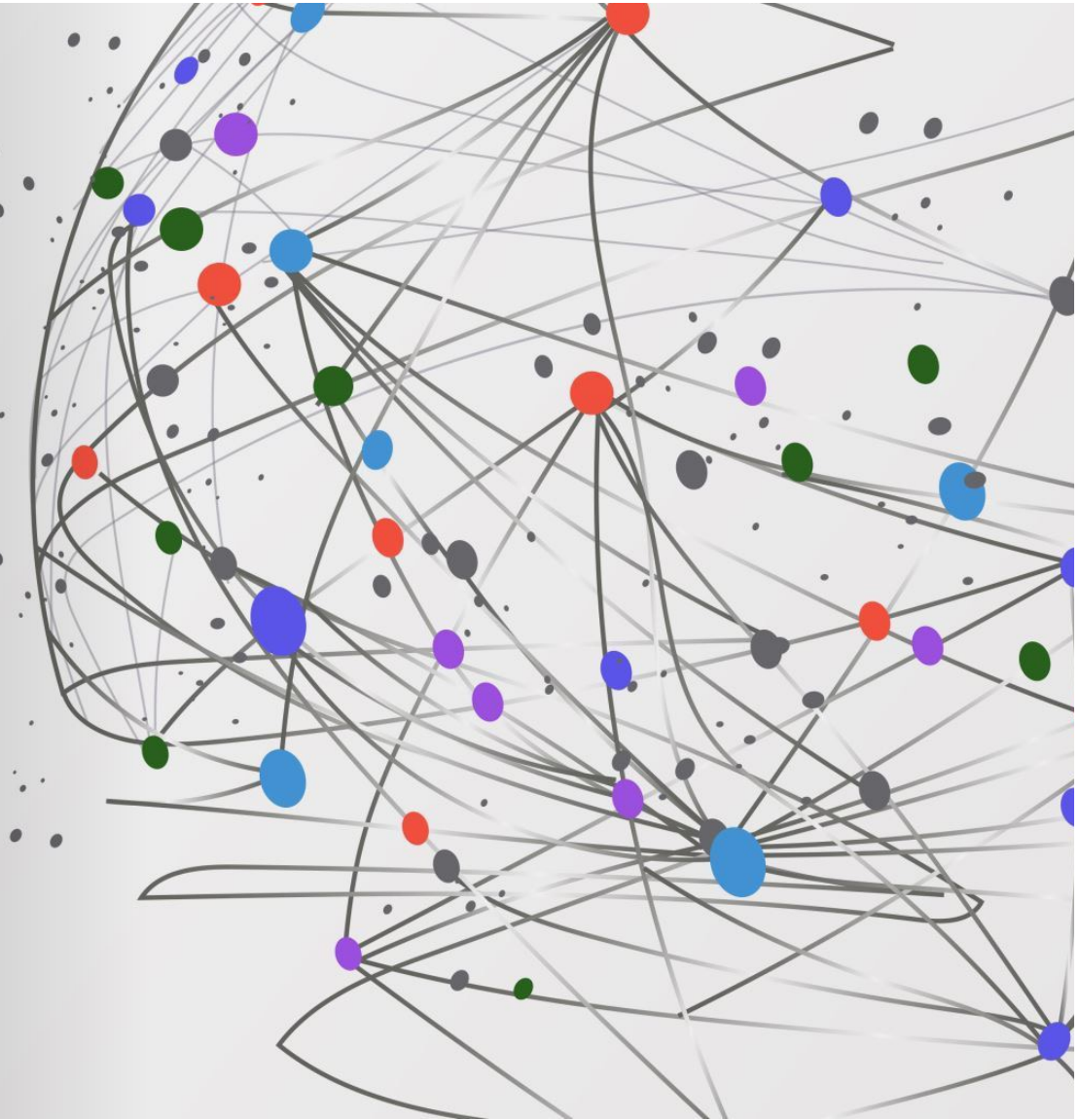
## Group 2 – FindIt

Engineering Design VII

Ryan Davis, Roy Tas, Cory Vitanza

Software Engineering & Engineering Management

School of Systems and Enterprises



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# Mission Statement

- **Mission:** Design and deliver a reliable, affordable, and user-friendly location tracking solution for families and caregivers of individuals at risk of wandering.
- **Goal:** Reduce stress by providing a dependable tool that enables quick and efficient tracking in critical situations.
- **Business Need:** Current lack of affordable long-range tracking technology for families.
- **Business Intent:** Bridge this gap by leveraging **LoRaWAN** for extended battery life and wide coverage.

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# Team Introduction

- Ryan Davis
  - 4/4 Software Engineering
- Roy Tas
  - 4/4 Engineering Management
- Cory Vitanza
  - 4/4 Software Engineering



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# Roles and Responsibilities

- Ryan Davis
  - Development Lead
  - Backend Development
  - User Advocate
  - System Administrator
- Roy Tas
  - Buildmeister
  - Infrastructure Development
  - Risk Management
  - Requirements Resource
- Cory Vitanza
  - Test Lead
  - Frontend Development
  - Architect
  - Designer

Primary Stakeholder :

- [Wearable Robotics Systems Lab](#)
- Dr. Damiano Zanutto

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## Key Drivers



### Enhanced Battery Life

LoRaWAN's low power consumption enables device to last much longer than competitors like AirTag.



### Technological Accessibility

Simple and user-friendly interface reduces stress when urgent tracking is needed



### Cost efficiency and subscription-free model

One-time purchase of the unit allows users to access tracking data without stressing about fees for tracking.

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# Key Constraints

- **Network Coverage**
  - LoRaWAN coverage in Urban areas allow for constant reliable GPS location data, however weak spots must be debugged.
- **Battery Life vs. Device Size**
  - Maintaining strong battery life while preserving battery size.
- **Privacy and Security**
  - Secure data handling, LoRaWAN data encryption, and user-controlled permissions are essential for user trust.
- **Inexpensive Deployment**
  - Affordable units is a primary selling point when considering competitors.
- **Interface Simplicity**
  - Diverse users require accessibility to be maintained when operating the user interface.

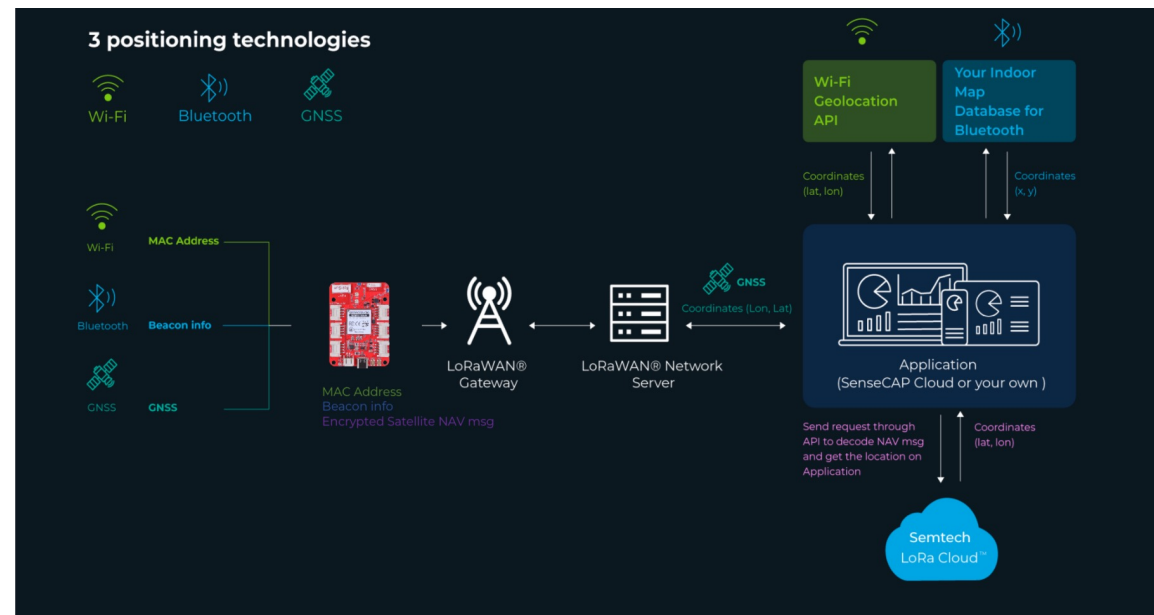
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# Risk Mitigation

- LoRaWAN Device Debugging
  - Current Issue: Limited to serial print statements; no button usage or re-flashing
  - Mitigation: Continue debugging + explore alternative devices (e.g., Raspberry Pi Nano)
- Tracking Algorithm Accuracy
  - Risk: Reduced precision in indoor/outdoor environments
  - Mitigation: Integrate both indoor & outdoor data sources for improved accuracy

# Development Plan – Hardware

- Wio Tracker 1110 Dev Board
- Semtech's [LR1110](#) LoRa® transceiver
- LoRaWAN Gateway (The Things Stack)

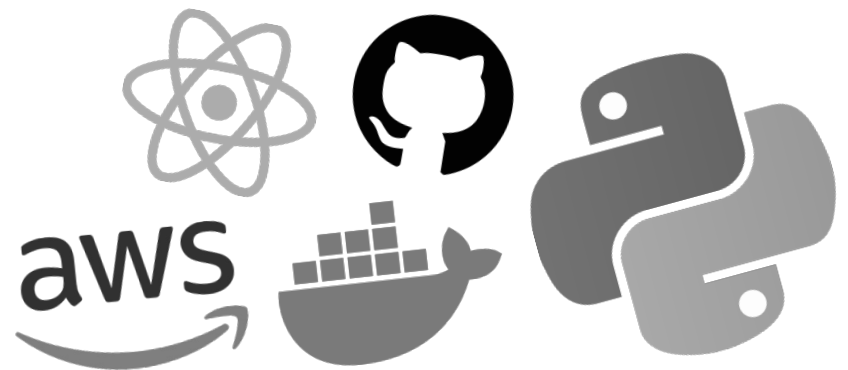


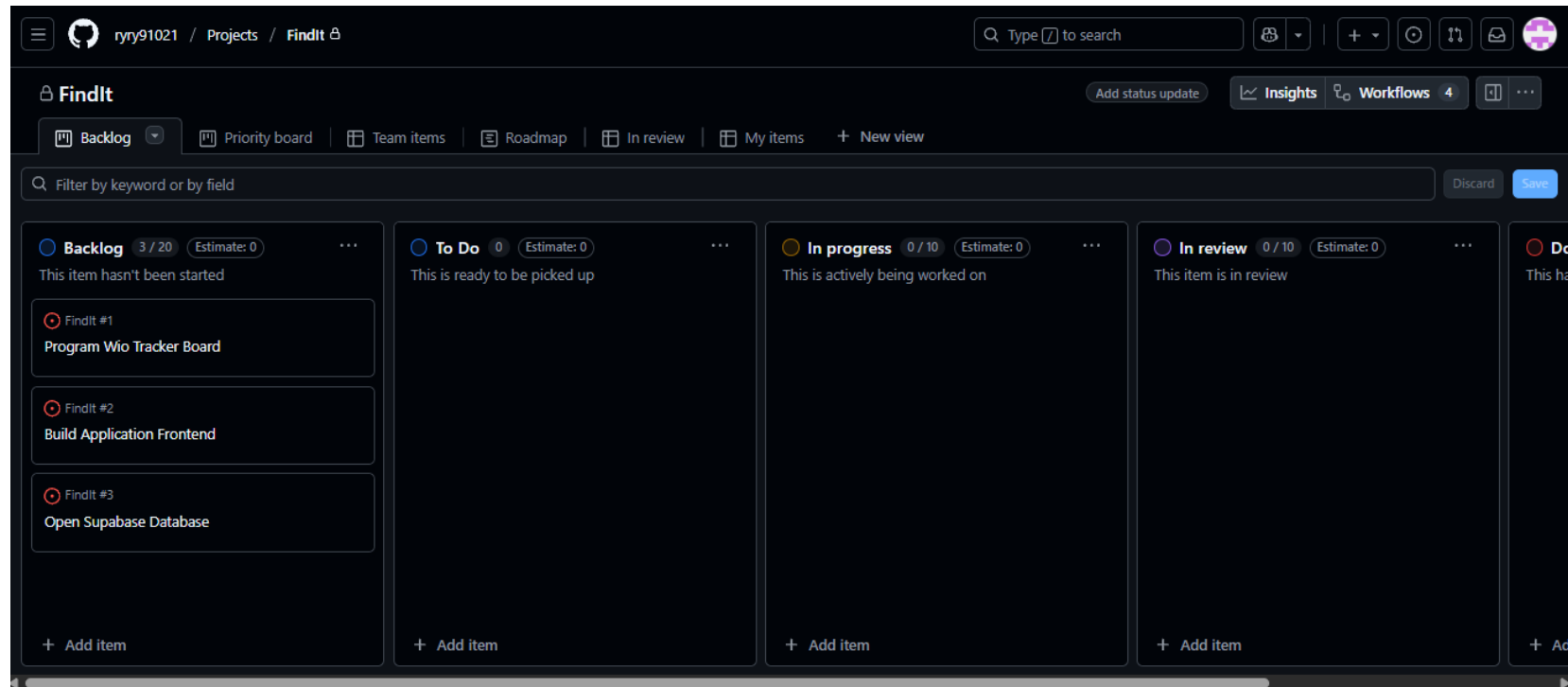


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# Development Plan - Software

- Backend Development
  - LoRaWAN protocol (TCIP)
  - Arduino IDE (C)
  - Supabase (PostgreSQL)
  - Python
    - Google Maps API
- Frontend Development
  - React (TypeScript)
  - Docker
  - AWS deployment
- Development Tools
  - GitHub
  - Kanban Board

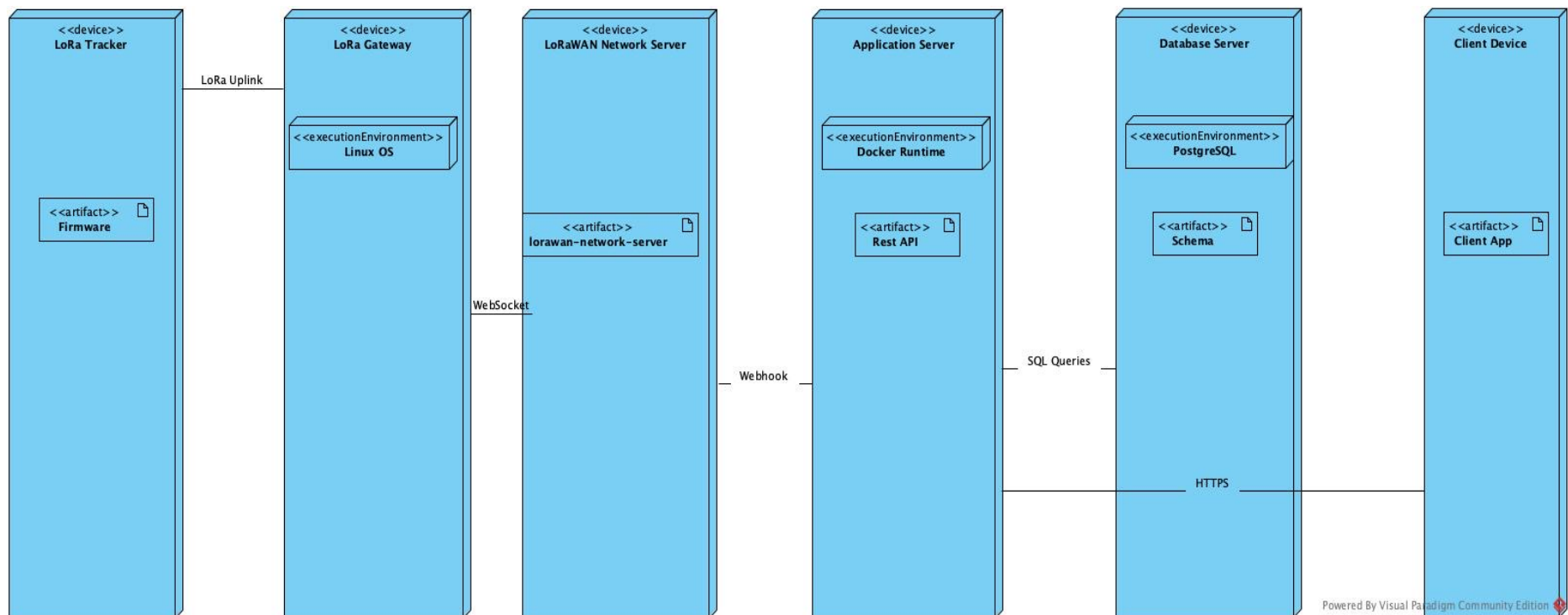




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# Agile Processes

# System Overview



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# User Stories

## PARENT OF A CHILD WITH AUTISM

- A parent of a child with autism expressed constant worry that their child might wander away at home or in public.
- They want a device that automatically alerts them when the child leaves a safe zone, so they can respond quickly without wasting time searching.

## CAREGIVER IN ASSISTED LIVING FACILITY

- A caregiver in an assisted living facility explained that they are responsible for multiple residents, many of whom are prone to wandering due to dementia.
- They need a system that can track multiple devices simultaneously and send alerts when a resident leaves a designated safe area, so they can act quickly.

## FAMILY MEMBER MONITORING BATTERY LIFE

- A family member stated that it is critical to know if the tracker's battery is running low.
- Without advance warning, the device could power off unexpectedly, leaving them unable to track their loved one during an emergency.

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# Use cases and Requirements

## Use Case

**UC1: Geofence Alert** - The system detects when the tracker exits a predefined boundary and sends an alert to the caregiver's phone.

**UC2: Multi-Device Monitoring** - A caregiver monitors multiple trackers simultaneously and receives notifications when any resident leaves a safe area.

**UC3: Battery Status Alert** - The tracker measures its battery life and sends a notification when power is low.

**UC4: Cost-Efficient Deployment** - The system operates without subscription costs, relying on LoRaWAN gateways, and supports multiple users.

## Mapped Requirements (Type)

R1: Geofencing with alerts (F)  
R2: Real-time location updates via mobile app (F)

R3: Multi-device monitoring (F)  
R4: Configurable notifications for multiple users/devices (F)

R5: Battery monitoring with alerts (F)  
R6: Push notifications for device status (F)

R7: Subscription-free operation (C)  
R8: Scalability for concurrent devices (N-F)

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# Glossary and Index Terms



## LoRaWAN

Long Range Wide  
Area Network



## LoRaWAN Gateway

Publicly accessible devices  
that forwards LoRa packets  
to LoRa servers (TTN)



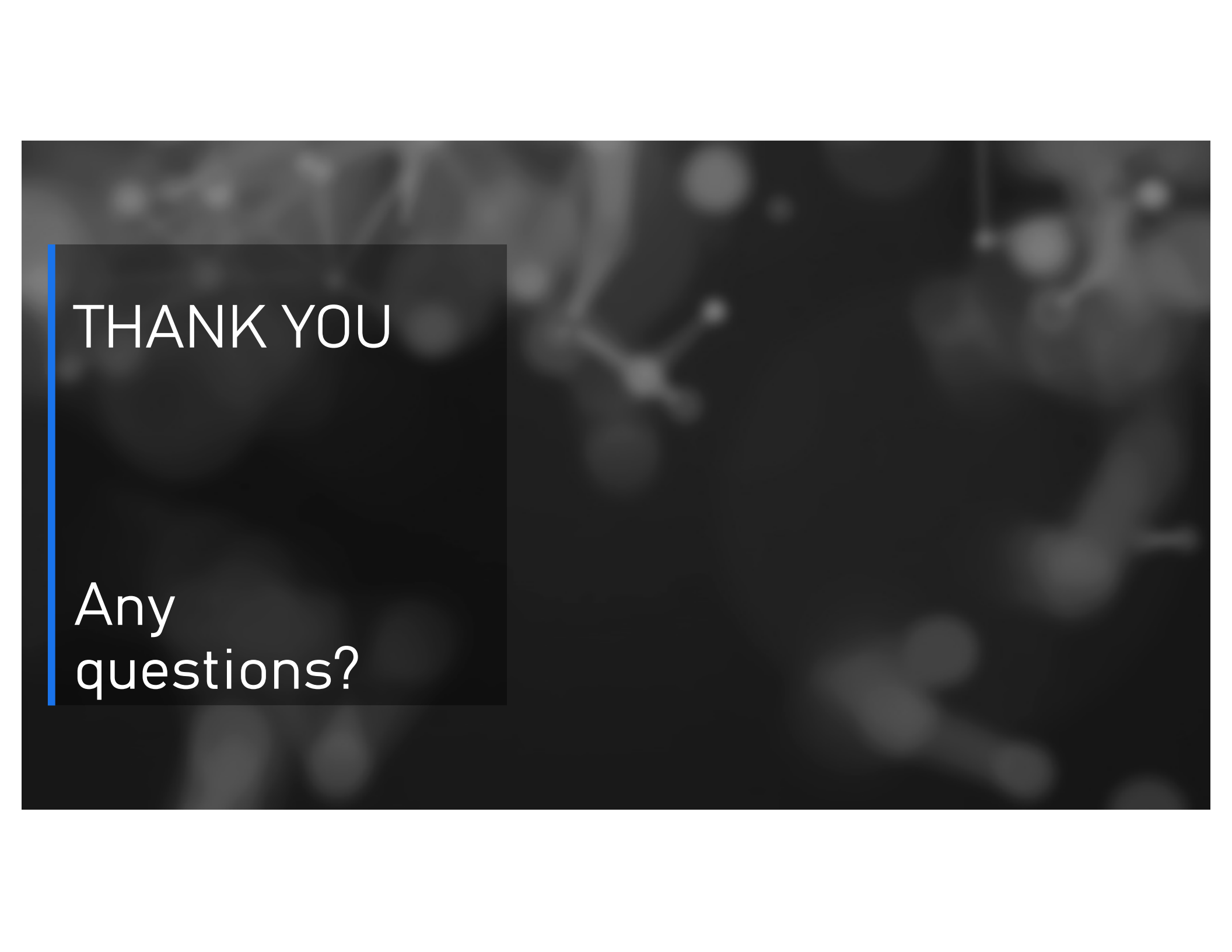
## The Things Network (TTN)

Open Source LoRaWAN  
community servers that allow  
for API data retrieval.

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# Citations

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THANK YOU

Any  
questions?